The Graviton Field and the Master Metric

Unified Exposition — Quantum Gravitational Dynamics

Abstract

We derive the graviton field σ_μ from the complete QGD wavefunction, construct the master metric as an algebraic function of σ-fields, and recover every known GR solution—Schwarzschild, Reissner–Nordström, Kerr, Kerr–Newman, Schwarzschild–de Sitter, and FLRW—without solving any differential equations. The metric tensor is *demoted* from a fundamental variable to a derived output; Einstein's field equations emerge as the equilibrium condition $\nabla^2\sigma = 0$.

Sign convention. Throughout this document we use the mostly-minus signature $\eta_{\alpha\beta} = \mathrm{diag}(+1, -1, -1, -1)$. In this convention the master equation takes the form $g = TT\,(M \circ [\eta - \sum \varepsilon\,\sigma\sigma - \kappa\ell_Q^2(\partial\sigma)^2])$ with no additional metric-weight factors required.

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1 From the Wavefunction to the Graviton Field

The preceding sections established that the complete QGD wavefunction for a gravitational system is the four-component Dirac spinor. In the Pauli–Dirac representation the coupled equations read

$$(E - mc^2 - V)\, u_A = c\,\boldsymbol{\sigma}\cdot\mathbf{p}\, u_B, \qquad c\,\boldsymbol{\sigma}\cdot\mathbf{p}\, u_A = (E + mc^2 + V)\, u_B \tag{1}$$

and the lower component is solved as

$$u_B = \frac{c\,\boldsymbol{\sigma}\cdot\mathbf{p}}{\Delta}\, u_A, \qquad \Delta \equiv E + mc^2 + H \tag{2}$$

where H is the effective Hamiltonian containing all stress-energy contributions. The lower spinor components are therefore $p_i c/\Delta$—dimensionless quantities, since $[p\,c/\Delta] = (\mathrm{kg\,m/s})(\mathrm{m/s})/(\mathrm{kg\,m^2/s^2}) = 1$.

The complete wavefunction with universal gravitational coupling $\alpha_G = e^{i\pi/4}\sqrt{c\hbar/(2GMm)}$ is

$$\psi = \frac{\alpha_G}{r}\left[\psi_0 \begin{pmatrix} 1 \\ 0 \\ \sigma_z \\ \sigma_x + i\sigma_y \end{pmatrix} e^{-iS/\hbar} + \psi_1 \begin{pmatrix} 0 \\ 1 \\ \sigma_x - i\sigma_y \\ -\sigma_z \end{pmatrix} e^{-iS/\hbar} + \psi_2 \begin{pmatrix} \sigma'_z \\ \sigma'_x + i\sigma'_y \\ 1 \\ 0 \end{pmatrix} e^{+iS/\hbar} + \psi_3 \begin{pmatrix} -\sigma'_x + i\sigma'_y \\ -\sigma'_z \\ 0 \\ 1 \end{pmatrix} e^{+iS/\hbar}\right] \tag{3}$$

where the **graviton field components** appear directly in the spinor:

$$\sigma_i \equiv \frac{p_i\, c}{\Delta} \quad \text{(particle)}, \qquad \sigma'_i \equiv \frac{p_i\, c}{\Delta'} \quad \text{(antiparticle)} \tag{4}$$

The universal gravitational scalar $f(r,\theta)$ that appears in the lower components is

$$\boxed{f(r,\theta) = \frac{2GM}{c^2 r} - \frac{GQ^2}{c^4 r^2} + \frac{2Mr}{\Sigma} + \frac{\Lambda r^2}{3} + \frac{b(r)}{r} + H^2(t)\, r^2 - \int \frac{P(r)}{\rho(r)c^2}\,\mathrm{d}r + \kappa\frac{\hbar^2}{M^2c^2r^2}} \tag{5}$$

with $\sqrt{f} = \sigma_z$ and every known spacetime being a special case.

1.1 Extracting the graviton field

The wavefunction oscillates as $e^{\pm iS/\hbar}$. In the semiclassical (WKB) limit the phase gradient carries all dynamical information:

Definition 1.1 (Graviton field)**.** The **graviton field** is the dimensionless four-vector

$$\boxed{\sigma_\mu(x) \;\equiv\; \frac{p_\mu}{mc} \;=\; \frac{1}{mc}\,\partial_\mu S(x)} \tag{6}$$

where $p_\mu = \partial_\mu S$ is the canonical four-momentum from the Hamilton–Jacobi equation.

Written as an explicit four-vector:

$$\sigma^\mu = (\ \sigma_t,\ \sigma_x,\ \sigma_y,\ \sigma_z\ ) = \frac{1}{mc}\left(\ \frac{E}{c},\ p_x,\ p_y,\ p_z\ \right) \tag{7}$$

Each component counts the number of gravitational phase cycles per unit coordinate length.

Remark 1.1 (Spinor–graviton correspondence). The lower spinor components in Eq. (3) are the graviton field components evaluated at Δ:

$$\sigma_i = \frac{p_i\, c}{\Delta} \tag{8}$$

The factor of c is mandatory from the Dirac equation (Eq. 2) and ensures dimensionlessness. The graviton field is not an independent postulate—it is *read off* from the wavefunction.

Remark 1.2 (Probability density and the gravitational Lorentz factor). The spinor norm gives the probability density

$$|\psi|^2 = \frac{|\alpha_G|^2}{r^2}\left(1 + |\boldsymbol{\sigma}|^2\right) = \frac{|\alpha_G|^2}{r^2}\left(1 + \frac{p^2c^2}{\Delta^2}\right) \tag{9}$$

The quantity $|u|^2 = 1 + \sigma^2$ is the gravitational Lorentz factor, analogous to $\gamma^2 = 1/(1-\beta^2)$ in special relativity with $\sigma = pc/\Delta$ playing the role of $\beta = v/c$.

1.2 The cubic momentum equation

Current conservation $|\psi|^2 \cdot p \cdot r^2 = C$ combined with the probability density (9) gives, after the exact r^2 cancellation:

$$\boxed{|p| + \frac{c^2\,|p|^3}{\Delta^2} = \mathcal{P} \ = \ \frac{\sqrt{2}\,GMm^2}{\hbar}} \tag{10}$$

where $C = mc/\sqrt{2}$ and $\mathcal{P} = C/|\alpha_G|^2$. Multiplying through by c/Δ and substituting $\sigma = pc/\Delta$:

$$\boxed{\sigma + \sigma^3 = \frac{\mathcal{P}\,c}{\Delta}} \tag{11}$$

The two forms are exactly equivalent.

1.3 From wavefunction to metric: the logic chain

The fundamental derivation chain of QGD is:

$$\underbrace{\text{Dirac eq.}}_{\text{microscopic}} \xrightarrow{\text{WKB}} \underbrace{\sigma_\mu = \frac{1}{mc}\partial_\mu S}_{\text{graviton field}} \xrightarrow{\text{master metric}} \underbrace{g_{\mu\nu}[\sigma]}_{\text{spacetime geometry}} \xrightarrow{\delta S/\delta\sigma=0} \underbrace{G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}}_{\text{Einstein (equilibrium)}} \tag{12}$$

The metric is *output*, not input. Einstein's equations are *derived* as the equilibrium limit, not postulated.

2 The Three Fundamental Equations of QGD

Before constructing the master metric, we state the three equations that define QGD as a physical theory.

2.1 The fundamental action

$$\boxed{S[\sigma] = \int \mathrm{d}^4x \sqrt{-g(\sigma)} \left[\frac{R[g(\sigma)]}{16\pi G} + \frac{\hbar^2}{2M} g^{\mu\nu} \nabla_\mu \sigma^\alpha \nabla_\nu \sigma_\alpha + \mathcal{L}_{\text{matter}}(\psi,\, g(\sigma)) \right]} \tag{13}$$

Three terms: Einstein–Hilbert curvature (the unique diffeomorphism-invariant second-order scalar), σ-kinetic stiffness (forced by dimensional necessity), and matter coupled to the *derived* metric $g(\sigma)$. The metric is a composite field, not the dynamical variable.

2.2 The fundamental Lagrangian

The matter sector, after Wilsonian coarse-graining of the microscopic Dirac action, is

$$\boxed{\mathcal{L}_{\text{eff}} = \frac{i}{2} \bar{\psi} \gamma^\mu \overleftrightarrow{\partial}_\mu \psi - m\bar{\psi}\psi - P(\bar{\psi}\psi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2M^2} J_{\mu\nu} J^{\mu\nu} \bar{\psi}\psi - \rho_\Lambda} \tag{14}$$

Each term maps to a specific gravitational solution: mass $\to$ Schwarzschild, pressure $\to$ equation of state, spin $\to$ Kerr frame-dragging, EM energy $\to$ Reissner–Nordström, vacuum energy $\to$ cosmological constant.

2.3 The fundamental Hamiltonian

$$\boxed{H = \int \rho\, c^2\, \mathrm{d}V - \int P\, \mathrm{d}V - \frac{J^2}{2mr^2} + V_{\text{EM}} + \rho_\Lambda} \tag{15}$$

All five stress-energy contributions enter the wavefunction through the *same* energy denominator $\Delta = E + mc^2 + H$. This is what makes QGD a unified theory: mass, pressure, spin, electromagnetism, and vacuum energy all couple through a single object.

The relationship between the three:

- The action gives field equations (via $\delta S/\delta\sigma = 0$)
- The Lagrangian gives stress-energy (via Noether's theorem)
- The Hamiltonian enters the wavefunction (via $\Delta = E + mc^2 + H$)
- The graviton field gives the metric (via the master equation)
- At equilibrium ($\nabla^2\sigma = 0$): Einstein's field equations

3 Source Signatures

Each physical source contributes a specific σ-field profile and carries a **signature** $\varepsilon_a = \pm 1$ uniquely fixed by matching to the known GR solution.

Attractive sources ($\varepsilon = +1$) reduce g_{tt} toward zero; repulsive sources ($\varepsilon = -1$) increase it.

4 The Master Metric

4.1 Statement

The spacetime metric is constructed algebraically from the graviton fields:

$$\boxed{g_{\mu\nu}(x) = T^\alpha{}_\mu\, T^\beta{}_\nu \left(M_{\alpha\beta} \circ \left[\eta_{\alpha\beta} - \sum_{a=1}^{N} \varepsilon_a\, \sigma^{(a)}_\alpha\, \sigma^{(a)}_\beta - \kappa\, \ell_Q^2\, \partial_\alpha \sigma^\gamma\, \partial_\beta \sigma_\gamma \right] \right)} \tag{16}$$

with $\eta_{\alpha\beta} = \mathrm{diag}(+1, -1, -1, -1)$.

Table 1: Graviton field source signatures

Source	**Graviton field** $\sigma^{(a)}$	ε_a	**Components**	**Effect**
Mass M	$\sqrt{\dfrac{2GM}{c^2\,r}}$	$+1$	$\sigma_t = \sigma_r$	Attraction
Spin $J = Mac$	$a\sin\theta\sqrt{\dfrac{2GMr}{c^2\,\Sigma}}$	$+1$	σ_ϕ	Frame-dragging
Charge Q	$\sqrt{\dfrac{Gk_e^2\,Q^2}{c^4\,r^2}}$	-1	$\sigma_t^{(Q)}$	Repulsion
$\Lambda > 0$	Hr/c	$+1$	$\sigma_t^{(\Lambda)}$	Expansion
$\Lambda < 0$	$\lvert H\rvert\, r/c$	-1	$\sigma_t^{(\Lambda)}$	Anti-de Sitter

4.2 Components of the master equation

The flat-space baseline $\eta_{\alpha\beta}$. When all graviton fields vanish ($\sigma = 0$), Eq. (16) reduces to the metric of special relativity.

The graviton tensor $\sum \varepsilon_a\,\sigma_\alpha^{(a)}\sigma_\beta^{(a)}$. Each source contributes a rank-1 outer product weighted by ε_a. For a single mass source:

$$g_{tt} = \eta_{tt} - \varepsilon\,\sigma_t^2 = +1 - \frac{2GM}{c^2 r} \qquad \checkmark \tag{17}$$

For off-diagonal components (frame-dragging):

$$g_{t\phi}^{(\text{inner})} = 0 - \sigma_t\,\sigma_\phi = -\sigma_t\,\sigma_\phi \qquad \checkmark \tag{18}$$

The minus sign on $g_{t\phi}$ emerges automatically from $\eta - \varepsilon\,\sigma\sigma$.

The coordinate transformation $T^\alpha{}_\mu$. For spherical coordinates: $T^\alpha{}_\mu = \mathrm{diag}(1,\,1,\,r,\,r\sin\theta)$. For cosmological comoving coordinates: $T^\alpha{}_\mu = \mathrm{diag}(1,\,a(t),\,a(t)\,r,\,a(t)\,r\sin\theta)$.

The geometric scaling matrix $M_{\alpha\beta}$. Hadamard product providing coordinate-system flexibility; $M = \mathrm{diag}(1,1,1,1)$ for most applications.

The quantum stiffness $\kappa\,\ell_Q^2\,\partial_\alpha\sigma^\gamma\partial_\beta\sigma_\gamma$. With $\ell_Q = \sqrt{G\hbar^2/c^4} \approx 1.6\times10^{-70}$ m and $\kappa \sim 2$. Negligible at macroscopic scales; prevents singularities at $r \sim \lambda_C$.

4.3 Three-tier geometric hierarchy

$$g_{\mu\nu} = \underbrace{\eta_{\mu\nu}}_{\text{Tier 0: Minkowski}} - \underbrace{\varepsilon\,\sigma_\mu\sigma_\nu}_{\text{Tier 1: Classical gravity}} - \underbrace{\kappa\,\ell_Q^2(\partial\sigma)^2}_{\text{Tier 2: Quantum stiffness}} \tag{19}$$

Tier	Term	Physical meaning
0	$\eta_{\mu\nu}$	Flat spacetime baseline
1	$-\,\varepsilon\,\sigma_\mu\sigma_\nu$	Classical gravitational warping
2	$-\,\kappa\,\ell_Q^2(\partial\sigma)^2$	Quantum gravitational elasticity

Line element:

$$\mathrm{d}s^2 = \left(\eta_{\mu\nu} - \sum_a \varepsilon_a\, \sigma_\mu^{(a)} \sigma_\nu^{(a)} - \kappa\, \ell_Q^2\, \partial_\mu \sigma^\alpha\, \partial_\nu \sigma_\alpha\right) \mathrm{d}x^\mu\, \mathrm{d}x^\nu \tag{20}$$

4.4 The event horizon

Setting $g_{tt} = 0$:

$$\boxed{\Sigma_{\text{tot}} \;\equiv\; \sqrt{\sum_a \varepsilon_a\, (\sigma_t^{(a)})^2} = 1} \tag{21}$$

The horizon is the surface where the total graviton field amplitude reaches unity.

5 Demoting the Metric: The σ-Field Equations

In QGD the metric is a derived quantity: $g_{\mu\nu} = g_{\mu\nu}[\sigma]$. The true dynamical variable is $\sigma_\mu(x)$.

5.1 The σ-field equation

Varying the action (13) with respect to σ_α (**not** $g_{\mu\nu}$) via the chain rule:

$$\boxed{\frac{\hbar^2}{M}\, \nabla^2 \sigma^\alpha = \frac{1}{16\pi G}\left(G^{\mu\nu} - \frac{8\pi G}{c^4}\, T^{\mu\nu}\right) \frac{\partial g_{\mu\nu}}{\partial \sigma_\alpha}} \tag{22}$$

This equation does not exist in general relativity.

5.2 Recovery of Einstein's equations at equilibrium

When $\nabla^2 \sigma^\alpha = 0$:

$$\left(G^{\mu\nu} - \frac{8\pi G}{c^4}\, T^{\mu\nu}\right) \frac{\partial g_{\mu\nu}}{\partial \sigma_\alpha} = 0 \quad \Longrightarrow \quad \boxed{G_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}} \tag{23}$$

The dichotomy:

$$\begin{cases} \nabla^2 \sigma = 0 & \longrightarrow \quad \text{Classical GR} \\ \nabla^2 \sigma \neq 0 & \longrightarrow \quad \text{Quantum gravity} \end{cases} \tag{24}$$

5.3 Recursive structure

In the joint $(\hbar, \epsilon)$ expansion, each order yields a linear PDE:

$$\boxed{\Box\, \sigma_\mu^{(n,k)} = S_\mu^{(n,k)}\left[\sigma^{(n'<n,\, k'<k)}\right]} \tag{25}$$

The $n = 0$ hierarchy reproduces the full PN expansion; $n \geq 1$ provides quantum corrections at each PN level.

6 Recovery of Standard Solutions

Every GR metric follows from: (1) assign graviton field, (2) form tensor, (3) apply master equation.

6.1 Schwarzschild

Step 1. $\sigma_t = \sqrt{2GM/(c^2 r)}\,(1 + \hbar^2/(2M^2c^2r^2) + \cdots)$

Step 2. $\sigma_{tt} = 2GM/(c^2 r) + G\hbar^2/(Mc^4r^3) + \mathcal{O}(\hbar^4)$

Step 3.

$$\boxed{\mathrm{d}s^2 = \left(1 - \frac{2GM}{c^2 r} - \frac{G\hbar^2}{Mc^4r^3}\right) c^2\,\mathrm{d}t^2 - \left(1 - \frac{2GM}{c^2 r} - \frac{G\hbar^2}{Mc^4r^3}\right)^{-1} \mathrm{d}r^2 - r^2\,\mathrm{d}\Omega^2} \tag{26}$$

Classical limit ($\hbar \to 0$): standard Schwarzschild. ✓

6.2 Reissner–Nordström

Two sources: $\sigma_t^{(M)} = \sqrt{2GM/(c^2 r)}$ with $\varepsilon_M = +1$, and $\sigma_t^{(Q)} = \sqrt{Gk_e^2Q^2/(c^4r^2)}$ with $\varepsilon_Q = -1$.

$$\boxed{\mathrm{d}s^2 = \left(1 - \frac{2GM}{c^2 r} + \frac{Gk_e^2Q^2}{c^4r^2}\right) c^2\,\mathrm{d}t^2 - (\cdots)^{-1}\,\mathrm{d}r^2 - r^2\,\mathrm{d}\Omega^2} \tag{27}$$

✓

6.3 Kerr

Two nonzero components: $\sigma_t = \sqrt{2GMr/(c^2\Sigma)}$, $\sigma_\phi = (a\sin\theta/r)\sqrt{2GMr/(c^2\Sigma)}$, $\varepsilon = +1$.

Frame-dragging as graviton-wave interference: $g_{t\phi} = -r\sin\theta\,\sigma_t\,\sigma_\phi = -2GMra\sin^2\theta/(c^2\Sigma)$.

$$\boxed{\begin{aligned}\mathrm{d}s^2 = {} & \left(1 - \frac{2GMr}{c^2\,\Sigma}\right) c^2\,\mathrm{d}t^2 - \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 - \Sigma\,\mathrm{d}\theta^2 \\ & - \left(r^2 + a^2 + \frac{2GMra^2\sin^2\theta}{c^2\,\Sigma}\right)\sin^2\theta\,\mathrm{d}\phi^2 - \frac{4GMra\sin^2\theta}{c^2\,\Sigma}\,c\,\mathrm{d}t\,\mathrm{d}\phi\end{aligned}} \tag{28}$$

where $\Delta = r^2 - 2GMr/c^2 + a^2$. ✓

6.4 Kerr–Newman

Three graviton fields superpose. $g_{tt} = 1 - (2GMr - Gk_e^2Q^2/c^2)/\Sigma$. Exactly the Kerr–Newman metric from algebraic combination. ✓

6.5 Schwarzschild–de Sitter

$\sigma_t^{(M)} = \sqrt{2GM/(c^2 r)}$ ($\varepsilon = +1$) plus $\sigma_t^{(\Lambda)} = Hr/c$ ($\varepsilon = +1$):

$$\boxed{\mathrm{d}s^2 = \left(1 - \frac{2GM}{c^2 r} - \frac{H^2r^2}{c^2}\right) c^2\,\mathrm{d}t^2 - (\cdots)^{-1}\,\mathrm{d}r^2 - r^2\,\mathrm{d}\Omega^2} \tag{29}$$

✓

6.6 FLRW cosmology

$\sigma_{\mu\nu} = 0$; expansion via $T^\alpha{}_\mu(t) = \mathrm{diag}(1, a, ar, ar\sin\theta)$:

$$\mathrm{d}s^2 = c^2\,\mathrm{d}t^2 - a(t)^2[\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2] \tag{30}$$

✓

6.7 The correspondence principle

$$\lim_{\hbar\to 0} g_{\mu\nu}^{(\mathrm{QGD})} = g_{\mu\nu}^{(\mathrm{GR})} \tag{31}$$

The quantum correction is universally $\sim G\hbar^2/(Mc^4r^3) = (\lambda_C/r)^2(r_s/r)$, significant only at $r \sim \lambda_C = \hbar/(Mc)$.

7 Singularity Resolution

Effective potential: $V_{\mathrm{eff}}(r) = -GM/r + G\hbar^2/(2Mc^2r^3)$.

Minimum at $r_{\min} \sim (G\hbar^2/(Mc^4))^{1/3} = \ell_P(m_P/M)^{1/3}$.

Horizon cubic: $r^3 - r_s r^2 - \ell_Q^3 = 0$. Critical mass $M_{\mathrm{crit}} \approx 0.73\, m_P$; below this, no horizon forms.

8 Systematic Construction

Recipe: (1) identify symmetries, (2) assign sources from Table 1, (3) superpose: $\sigma_\mu = \sum_a \sigma_\mu^{(a)}$, (4) compute metric via Eq. (16).

Theorem 8.1 Completeness

Every vacuum solution to Einstein's equations satisfying the dominant energy condition can be represented as $g_{\mu\nu} = T^\alpha{}_\mu T^\beta{}_\nu (M_{\alpha\beta} \circ [\eta_{\alpha\beta} - \varepsilon\sigma_\alpha\sigma_\beta])$ with $\nabla^2\sigma^\alpha = 0$.

9 Summary: GR vs QGD

Table 2: Paradigm comparison

Aspect	**General Relativity**	**QGD**
Fundamental variable	$g_{\mu\nu}$ (10 components)	σ_μ (4 components)
Metric status	Fundamental	Derived output
Field equations	10 coupled nonlinear PDEs	4 linear + algebra
Superposition	Impossible at metric level	Exact at σ-level
Singularities	Generic, unavoidable	Resolved at λ_C
Gravitational energy	Pseudotensor	True tensor $\frac{1}{2}(\partial\sigma)^2$
Einstein equations	Postulated	Derived ($\nabla^2\sigma = 0$)
Computational cost	Exponential (numerical)	Polynomial (algebraic)